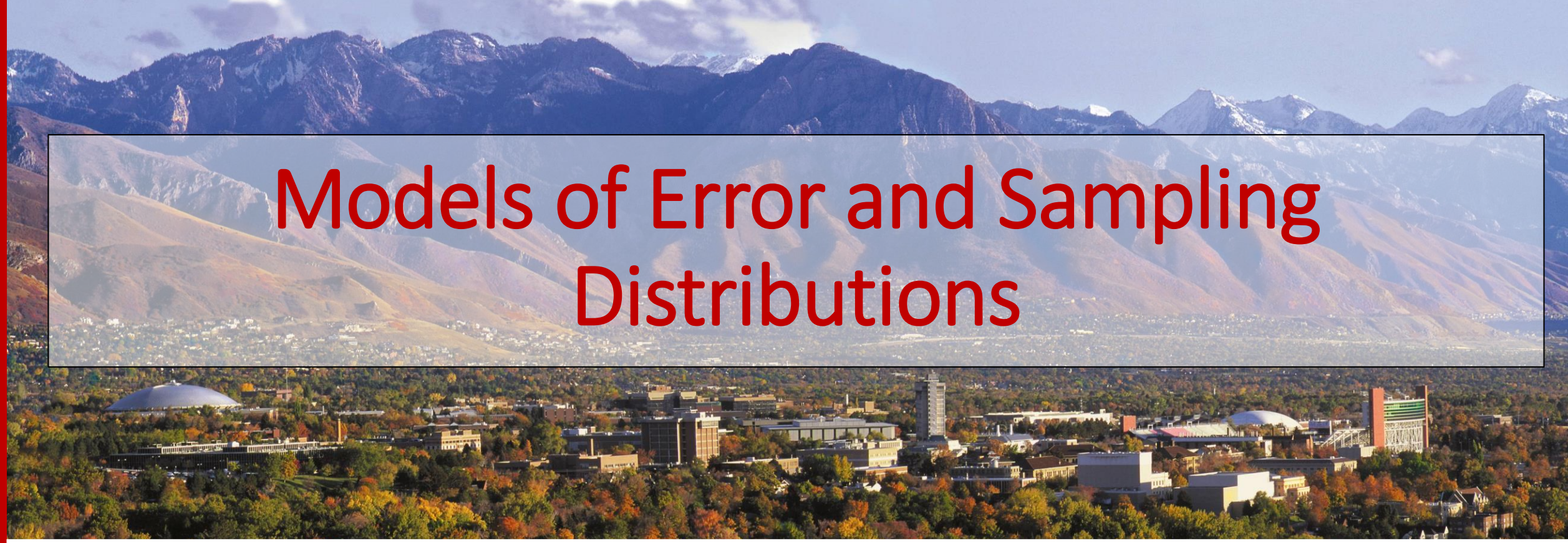

Models of Error and Sampling Distributions

Keith Lohse, PhD

Assistant Professor

Department of Health, Kinesiology, and Recreation

Department of Physical Therapy and Athletic Training

University of Utah

keith.lohse@health.utah.edu

@keith_lohse

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Estimation and sampling

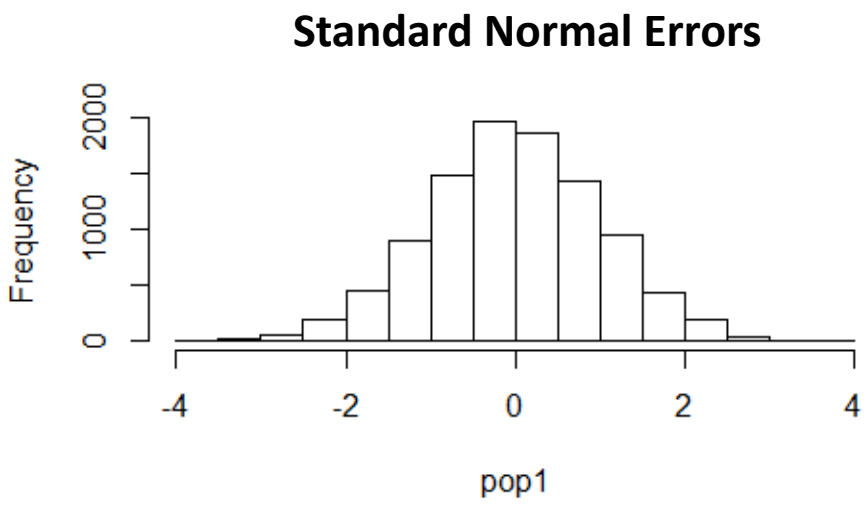
- We are often trying to make ***inferences*** about a ***population*** from a ***sample***.

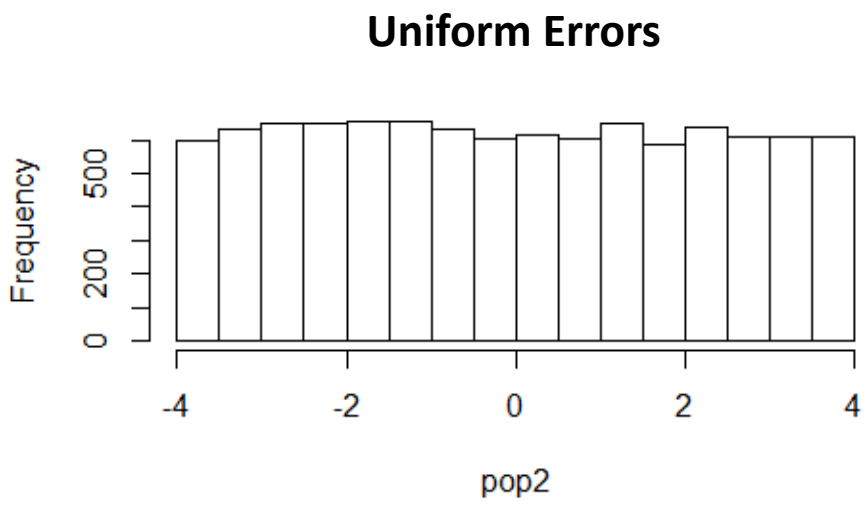
- Values we observe in our sample are ***estimates*** of "true" values in the population (***estimands***).
  - The function used to obtain the value of the estimate is the ***estimator***.
  - Thus, we want to employ functions (estimators) that give us representative values (estimates) for the quantity of interest (estimands).

Desirable Properties of Estimators

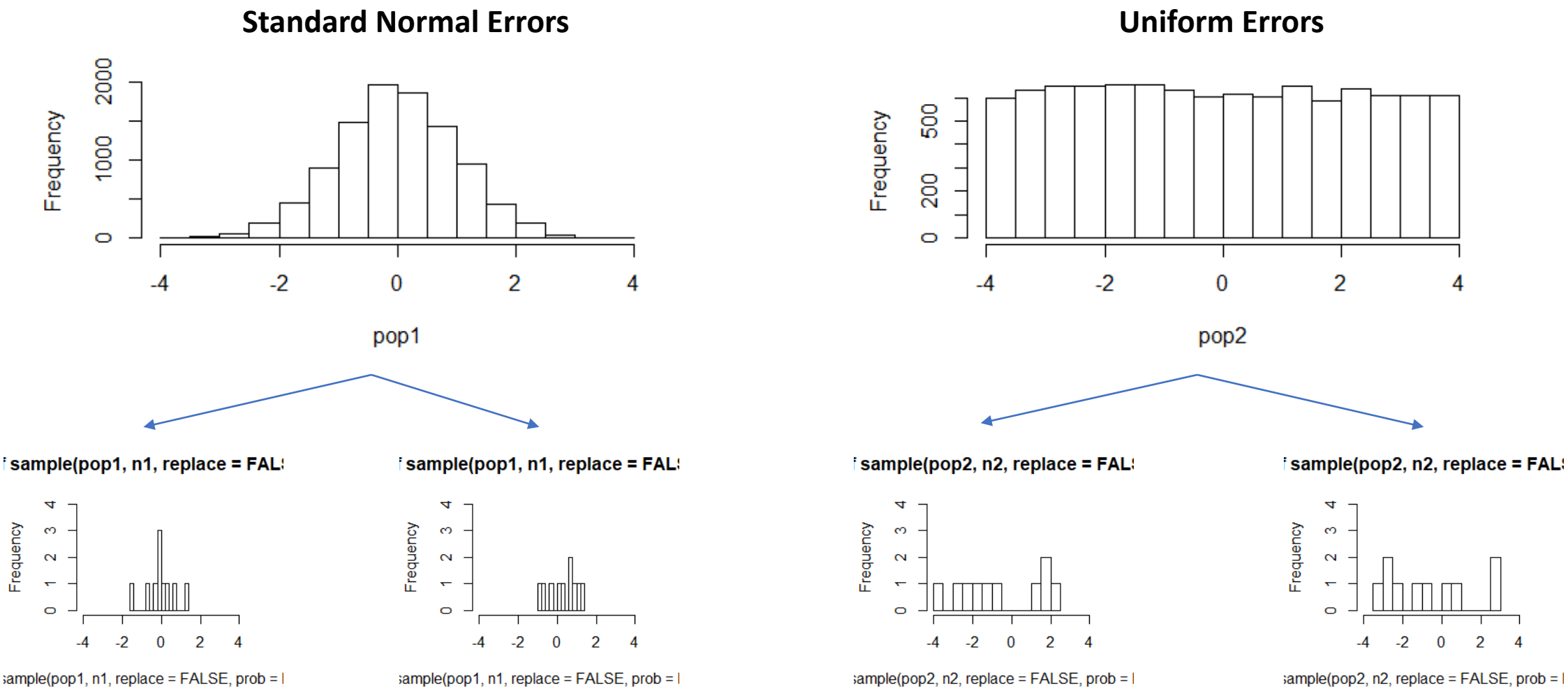
- ***Unbiased***
  - The estimator is not necessarily accurate, but it is as likely to over-estimate as it is to under-estimate.
- ***Efficient***
  - Given that two estimators are unbiased, we will prefer the one that has a smaller variance (or spread) around the estimand.
- ***Consistent***
  - As the sample size increases, the estimator produces better and better approximations of the estimand.

Sampling from a population of errors





Sampling from a population of errors



Across 10k samples of n = 10

	Mean(Mean)	SD(mean)	Mean(Median)	SD(Median)
“Normal” Population	-0.007	0.320	-0.008	0.373
Uniform Population	-0.025	0.722	-0.049	1.097

Across 10k samples of n = 10

	Mean(Means)	SD(Means)	Mean(Medians)	SD(Medians)
“Normal” Population	-0.007	0.320	-0.008	0.373
Uniform Population	-0.025	0.722	-0.049	1.097

Across 10k samples of n = 50

	Mean(Means)	SD(means)	Mean(Medians)	SD(Medians)
“Normal” Population	-0.005	0.144	-0.012	0.173
Uniform Population	-0.037	0.327	-0.073	0.555

Choosing among different estimators

- Both the mean and the median are ***unbiased*** and ***consistent*** estimators.

- In the examples we looked at, ***the mean tends to be a more efficient estimator than the median***.

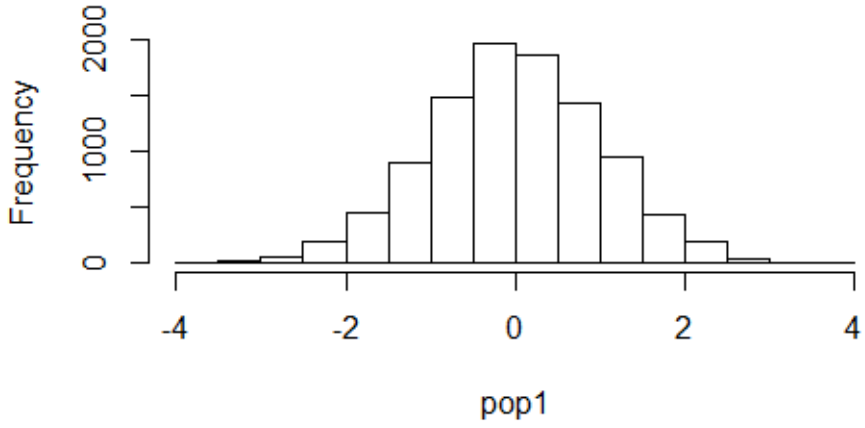
- Thus, we would choose the sum of squared errors and it's associated estimator (i.e., the mean) as the preferred method.
  - However, this conclusion will depend on the underlying distribution.

Choosing among different estimators

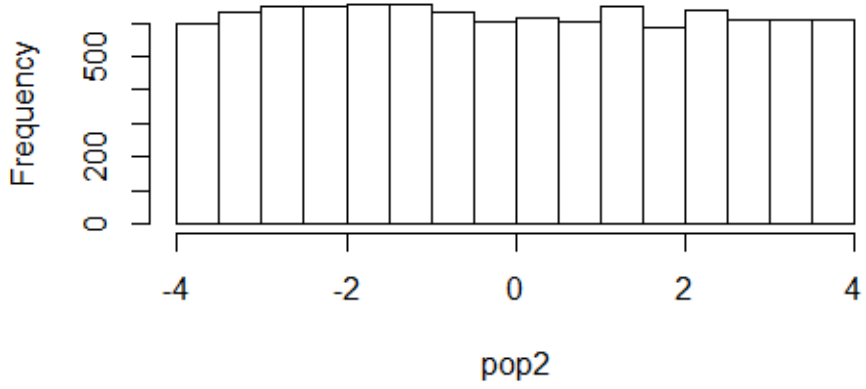
- There are many possible distributions of errors that we might assume, but there are good reasons for choosing the normal distribution.
  - Empirical data show us that (approximately) normal distributions are very common.
  - Linear combinations of random variables (even non-normal variables!) lead to (approximately) normal distributions.
  - The ***central limit theorem*** shows that sample means, even from non-normal distributions will be approximately normal.
  - Even when non-normal, many distributions can be transformed to become approximately normal.
    - E.g., skewed distributions can be square-root transformed or log transformed.

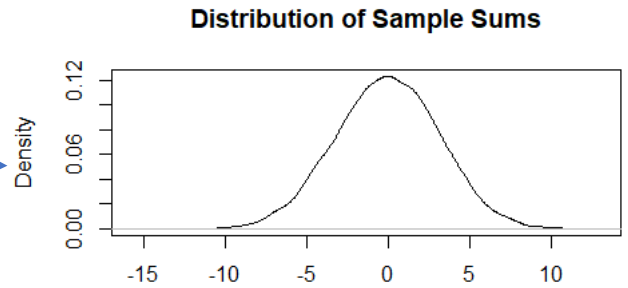
Distribution of Sample Sums

Standard Normal Errors

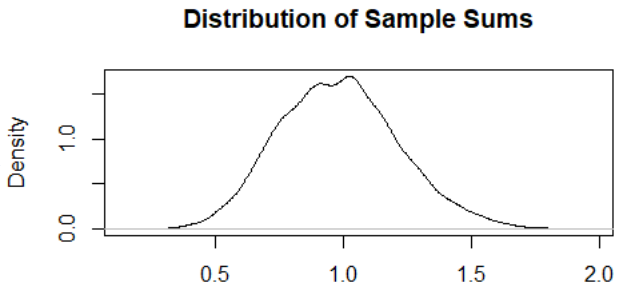


Uniform Errors





10k samples of n = 10



10k samples of n = 10